Shahabuddin Ahmed

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PROFESSIONAL SUMMARY

Senior Backend Software Engineer with 7 years of experience designing scalable, reliable, and high-performance backend systems using Node.js, NestJS, TypeScript, RabbitMQ, Kafka, Redis, MySQL, and Kubernetes. Strong expertise in microservices, distributed systems, asynchronous messaging, and performance optimization. Led the migration from a legacy Sails.js monolith to modern microservices, improving scalability by 30% and reducing API latency by 40%. Skilled in system design, API architecture, caching, observability, and mentoring backend engineers. Currently building AI coding-agent infrastructure, sandboxed code execution, and workflow automation integrations for an AI-powered website builder.

EXPERIENCE

AppifyLab Dhaka, Bangladesh
Senior Software Engineer *Sep 2026 – Present*

- Building the AI coding agent for EzyStudio, a multi-tenant AI website builder. The **agent harness** runs on the Vercel AI SDK and OpenRouter, with tool calling, run-state management, verification loops, model selection, prompt caching, and per-run cost tracking.
- Building **sandboxed code execution on E2B**, with workspace files and snapshots stored in Cloudflare R2, sandbox-to-storage file sync, and reliable dev-server startup for live previews.
- Building a **React-based automation workflow builder** with webhook integrations for Zapier and Pabbly, enabling customers to trigger and connect third-party workflows.

ShareTrip Ltd. Dhaka, Bangladesh
Senior Backend Software Engineer *Jul 2019 – Aug 2026*

- Led the migration from a legacy **Sails.js monolith** to a distributed NestJS microservices architecture, improving scalability by 30% and reducing API latency by 40%.
- Replaced the legacy synchronous flight search with an **asynchronous RabbitMQ + Redis pipeline**, scaling search capacity from 5K to 25K+ flights per request and eliminating provider bottlenecks.
- Protected high-cost flight search APIs from scraping and unauthorized clients using **app attestation** (Google Play Integrity on Android, Apple App Attest on iOS), **single-use short-lived tokens** issued server-side for the web, and **Cloudflare bot protection**.
- Modernized hotel search by migrating from ElasticSearch to a cost-efficient **TypeSense + MongoDB** architecture with intelligent fallback, enabling sub-30s progressive search across 2.2M+ properties and 10K+ daily searches.
- Rewrote the notification microservice with an **event-driven RabbitMQ** architecture for email, SMS, and push, processing 12K+ notifications daily (400K+/month) with dynamic Puppeteer-generated attachments.
- Rebuilt logging for sensitive flight data as a **queue-based asynchronous service** with structured logs and S3 archival, and strengthened observability with **Grafana dashboards**, metrics, and alerting.
- **Managed and mentored backend engineers**, leading architecture reviews and driving best practices in microservices, event-driven design, and distributed systems.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Go, Java
- **Backend:** Node.js, NestJS, AdonisJS, Express, Sails.js
- **AI & Agents:** Vercel AI SDK, OpenRouter, E2B, LLM tool calling, prompt caching
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, TypeSense
- **Messaging:** RabbitMQ, Kafka, gRPC
- **DevOps & Cloud:** Docker, Kubernetes, AWS (EC2, S3, RDS, Lambda), Cloudflare R2, Nginx, Grafana
- **Frontend:** React, Next.js, Angular, Redux Toolkit
- **Practices:** API Security, Microservices, Event-Driven Architecture, TDD, DDD, CI/CD, System Design

EDUCATION

University of Rajshahi Rajshahi, Bangladesh
B.Sc. (Engg.) in Computer Science & Engineering *Mar 2014 – Mar 2019*